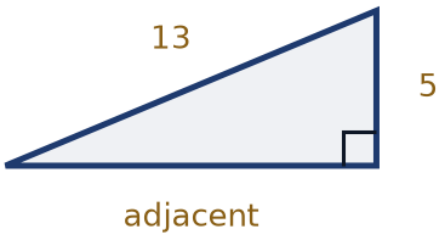
Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Right-triangle trigonometric ratios

In a right triangle, angle θ has an opposite side of 5 and hypotenuse 13. Find $\sin\theta$ and the length of the adjacent side.



2 [Foundational] • Target: Right-triangle trigonometric ratios

In a right triangle, angle θ has an opposite side of 15 and hypotenuse 17. Find $\cos\theta$.

3 [Foundational] • Target: Right-triangle trigonometric ratios

In a right triangle, angle θ has an adjacent side of 7 and hypotenuse 25. Find $\sin\theta$ and $\tan\theta$.

4 [Foundational] • Target: Right-triangle trigonometric ratios

In a right triangle, the two legs are 20 and 21, and the hypotenuse is 29. Find $\sin\theta$ and $\cos\theta$ for the angle θ opposite the side of length 20.

5 [Foundational] • Target: Special angles

State the exact value of $\sin(30^\circ)$.

6 [Foundational] • Target: Special angles

State the exact values of $\sin(90^\circ)$ and $\cos(0^\circ)$.

7 [Foundational] • Target: Converting between radians and degrees

Convert 180° to radians.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Converting between radians and degrees

Convert $\frac{5\pi}{6}$ radians to degrees.

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9 [Intermediate] • Target: The Pythagorean identity

Given $\sin\theta = \frac{3}{5}$ with θ acute, find $\cos\theta$ using $\sin^2\theta + \cos^2\theta = 1$.

10 [Intermediate] • Target: The Pythagorean identity

Given $\cos\theta = \frac{5}{13}$ with θ acute, find $\sin\theta$.

11 [Intermediate] • Target: Graphs: amplitude and period

State the amplitude and period of $y = 5\sin(x)$.

12 [Intermediate] • Target: Graphs: amplitude and period

A sine graph has amplitude 4 and period π. Write its equation in the form $y = A\sin(Bx)$.

13 [Intermediate] • Target: Graphs: phase shift and vertical shift

The graph of $y = \sin(x)$ is shifted $\frac{\pi}{3}$ to the right and 2 units up. Write the new equation.

14 [Intermediate] • Target: Graphs: phase shift and vertical shift

State the vertical shift and midline of $y = 3\cos(x) + 5$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Graphs: phase shift and vertical shift • Cross-Lesson: 7.01 Integration & Applications

The graph of $y = \cos(x)$ is shifted $\frac{\pi}{4}$ to the left. Write the new equation.

16 [Challenging] • Target: Reference angles in other quadrants • Cross-Lesson: 8.01 Exponentials & Logarithms

Find $\sin(150^\circ)$ using the reference angle.

17 [Challenging] • Target: Complementary angle relationships • Cross-Lesson: 9.01 Statistics & Probability

Given $\sin(20^\circ) \approx 0.342$, use the relationship $\sin\theta = \cos(90^\circ - \theta)$ to find $\cos(70^\circ)$.

18 [Challenging] • Target: The unit circle: coordinates of key points • Cross-Lesson: 1.01 Algebra & Equations

State the coordinates $(\cos\theta, \sin\theta)$ of the point on the unit circle at $\theta = 60^\circ$.

19 [Challenging] • Target: Word problems: angles of elevation • Cross-Lesson: 2.01 Functions & Transformations

This question has two parts. A drone hovers directly above a landmark. From a point 200 meters from the base of the landmark, the angle of elevation to the drone is 35°.

a Write an equation relating the drone's height h to the given information, using the tangent ratio.

b Find h, to the nearest meter, given $\tan(35^\circ) \approx 0.700$.

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20 [Challenging] • Target: Word problems: angles of elevation • Cross-Lesson: 3.01 Sequences & Series

This question has two parts. A ladder 10 meters long leans against a wall, making an angle of 65° with the ground.

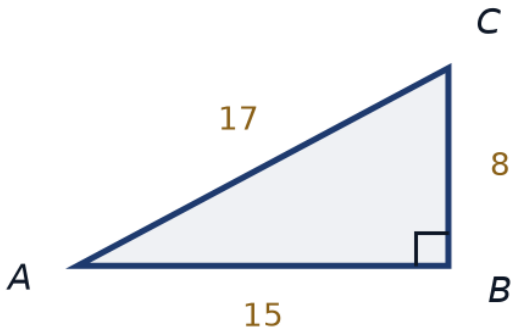
a Write an equation for the height h the ladder reaches up the wall, using the sine ratio.

b Find h, to the nearest tenth of a meter, given $\sin(65^\circ) \approx 0.906$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Right-triangle trigonometric ratios

In the right triangle shown, angle θ has opposite side 8 and hypotenuse 17. Find $\sin\theta$.



a. $\frac{17}{8}$

b. $\frac{8}{15}$

c. $\frac{15}{17}$

d. $\frac{8}{17}$

22 [Foundational] • Target: MCQ — Right-triangle trigonometric ratios

In a right triangle, angle θ has opposite side 9 and hypotenuse 41. Find $\cos\theta$.

a. $\frac{9}{41}$

b. $\frac{9}{40}$

c. $\frac{41}{40}$

d. $\frac{40}{41}$

23 [Foundational] • Target: MCQ — Right-triangle trigonometric ratios

In a right triangle, angle θ has adjacent side 20 and hypotenuse 29. Find $\sin\theta$.

a. $\frac{21}{20}$

b. $\frac{20}{29}$

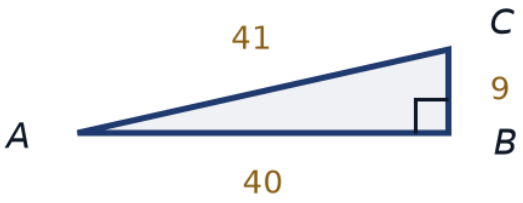
c. $\frac{21}{29}$

d. $\frac{29}{21}$

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24 [Foundational] • Target: MCQ — Right-triangle trigonometric ratios

In the right triangle shown, the two legs are 9 and 40, and the hypotenuse is 41. Find $\sin\theta$ and $\cos\theta$ (where θ is opposite the side of length 9).



a. $\sin\theta = \frac{9}{40}, \cos\theta = \frac{40}{9}$

b. $\sin\theta = \frac{40}{41}, \cos\theta = \frac{9}{41}$

c. $\sin\theta = \frac{41}{9}, \cos\theta = \frac{41}{40}$

d. $\sin\theta = \frac{9}{41}, \cos\theta = \frac{40}{41}$

25 [Foundational] • Target: MCQ — Right-triangle trigonometric ratios

In a right triangle, angle θ has opposite side 7 and adjacent side 24. Find $\tan\theta$.

a. $\frac{24}{7}$

b. $\frac{24}{25}$

c. $\frac{7}{25}$

d. $\frac{7}{24}$

26 [Foundational] • Target: MCQ — Graphs: amplitude and period

A cosine graph has amplitude 3 and period 4π. Write its equation in the form $y = A\cos(Bx)$.

a. $y = 3\cos(4\pi x)$

b. $y = 4\pi\cos(3x)$

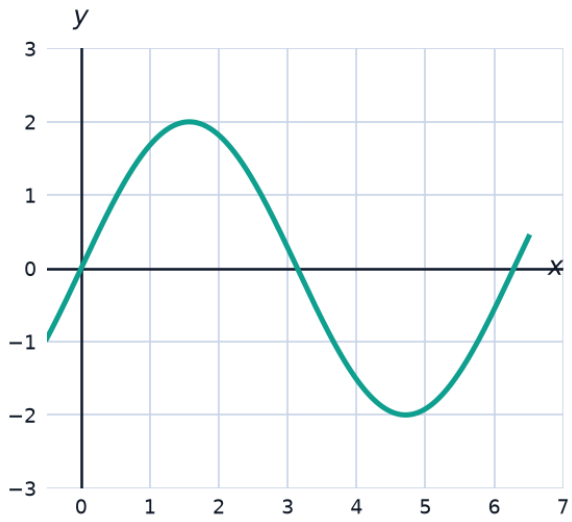
c. $y = 3\cos\left(\frac{x}{2}\right)$

d. $y = 3\cos(2x)$

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27 [Foundational] • Target: MCQ — Graphs: amplitude and period

The graph shows $y = A\sin(x)$ for some A. Find A by reading the graph's amplitude.



a. $A = 1$

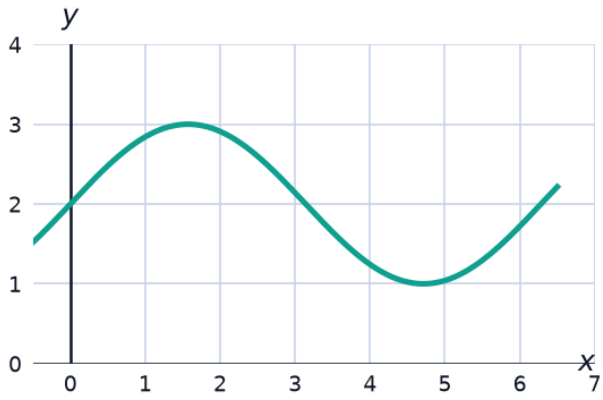
b. $A = 2\pi$

c. $A = 4$

d. $A = 2$

28 [Foundational] • Target: MCQ — Graphs: phase shift and vertical shift

The graph shows $y = \sin(x)$ shifted vertically. Find the new midline by reading the graph.



a. $y = 0$

b. $y = 2$

c. $y = 1$

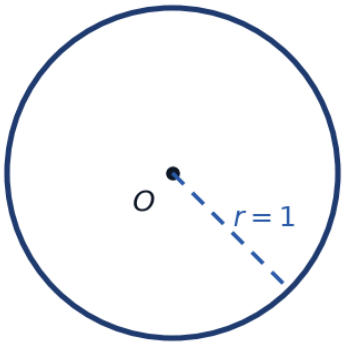
d. $y = -2$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: MCQ — The unit circle: coordinates of key points**

The unit circle shown marks a point at angle $\theta = 60^\circ$. What are its coordinates?



a. $\left(\frac{\sqrt{3}}{2}, 0.5\right)$

b. $(0.5, 0.5)$

c. $(1, 0)$

d. $\left(0.5, \frac{\sqrt{3}}{2}\right)$

30 **[Intermediate]** • **Target: MCQ — Word problems: angles of elevation**

A drone hovers directly above a landmark. From a point 150 meters from the landmark's base, the angle of elevation to the drone is 40°. Find the drone's height, to the nearest meter.

a. 150

b. 196

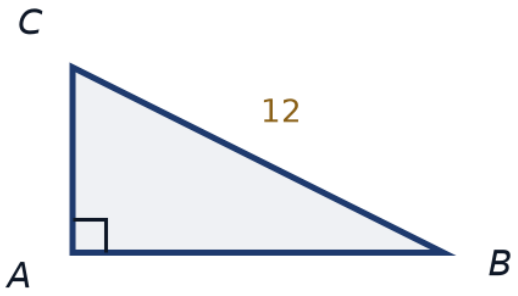
c. 126

d. 115

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31 **[Intermediate]** • **Target: MCQ — Word problems: angles of elevation**

A ladder leans against a wall, making an angle of 70° with the ground. If the ladder shown is 12 meters long, find how high up the wall it reaches, to the nearest meter.



a. 4

b. 12

c. 8

d. 11

32 **[Intermediate]** • **Target: MCQ — Word problems: angles of elevation**

From the top of an 80 m cliff, the angle of depression to a boat is 25°. Find the horizontal distance from the base of the cliff to the boat, to the nearest meter.

a. 37

b. 80

c. 172

d. 189

33 **[Intermediate]** • **Target: MCQ — Right-triangle trigonometric ratios**

In a right triangle, angle θ has opposite side 18, adjacent side 80, and hypotenuse 82. Find $\sin\theta$.

a. $\frac{41}{9}$

b. $\frac{3}{14}$

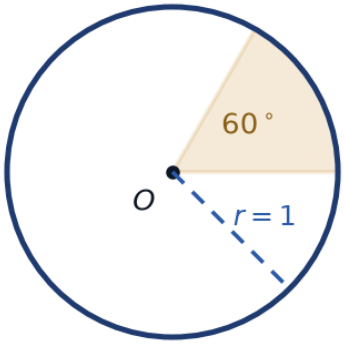
c. $\frac{8}{41}$

d. $\frac{9}{41}$

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34 **[Intermediate]** • **Target: MCQ — Special angles**

[redacted] $\theta = \frac{\pi}{3}$ [redacted] In the unit circle, what are the Cartesian coordinates of the point corresponding to an angle $\theta = \frac{\pi}{3}$ radians?



a. $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

b. $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

c. $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

d. $(0, 1)$

35 **[Intermediate]** • **Target: MCQ — Converting between radians and degrees**

[redacted] ABC[redacted] [redacted] [redacted] B [redacted] 90 [redacted] [redacted] [redacted] [redacted] AB [redacted] 3 [redacted] [redacted] [redacted] BC [redacted] 4 [redacted] [redacted] [redacted] [redacted] [redacted] AC[redacted] In a right-angled triangle ABC, where the angle at B is 90 degrees, if the length of side AB is 3 cm and the length of side BC is 4 cm, what is the length of the hypotenuse AC?

a. 5 [redacted]

b. 6 [redacted]

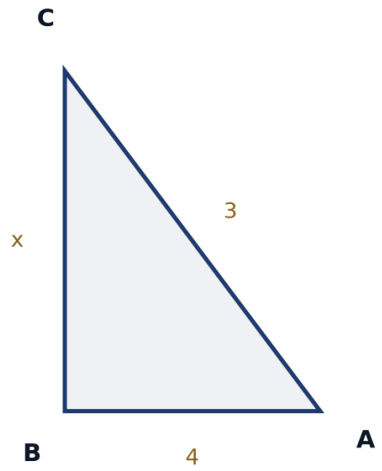
c. 7 [redacted]

d. 8 [redacted]

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36 [Intermediate] • Target: MCQ — The Pythagorean identity

■■ ■■■■■■■ ■■■■■■■ ■■■■■■■■ ABC■ ■■■■ ■■■■■■■■ ■■■■■■■■ ■■■■ B■ ■■■■ ■■■■ ■■■■ ■■■■■■ AB ■■■■■■ 3 ■■ ■■■■■ ■■■■■■ BC ■■■■■■ 4 ■■■■ ■■■■ ■■ ■■■■ ■■■■■■ AC■ In a right-angled triangle ABC, with the right angle at B, if the length of side AB is 3 cm and the length of side BC is 4 cm, what is the length of the hypotenuse AC?



a. 5 ■■

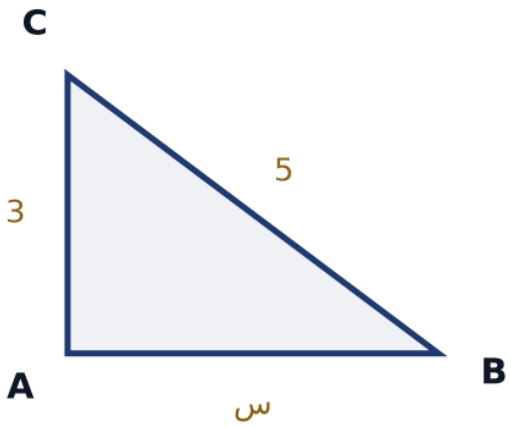
b. 7 ■■

c. 25 ■■

d. 12 ■■

37 [Intermediate] • Target: MCQ — Simplifying trigonometric expressions

■■ ■■■■■ ■■■■■ ■■■■■■■■■ ■■■■■■ $= 5$ ■■■■■ ■■■■■■■■ $= 3.$ ■■ ■■■■ ■■■■■■ ■■■■■■■ In a right triangle, the hypotenuse is 5 and one leg is 3. Find the other leg.



a. 6

b. 1

c. 3

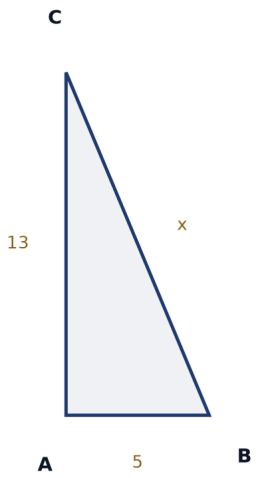
d. 4

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Graphs: amplitude and period • Cross-Lesson: 1.01 Algebra & Equations

■■ ■■■■■■■ ■■■■■■■ ■■■■■■■■ ■■■■■■■■ ■■ ■■ ■■■■ ■■■■■■ x■ In the right-angled triangle shown, what is the length of side x?



a. 10

b. 12

c. 14

d. 16

39 [Challenging] • Target: MCQ — Graphs: phase shift and vertical shift • Cross-Lesson: 2.01 Functions & Transformations

Consider a point $P(\cos\theta, \sin\theta)$ on the unit circle in the Cartesian plane. If the distance from the origin to the point $Q(x, y)$ is 5, and the angle between the positive x-axis and the line segment OQ is α, such that $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. If $x = \cos\alpha$ and $y = \sin\alpha$, find the maximum possible value of the product of the x-coordinates of P and Q, i.e., $\cos\theta \cdot \cos\alpha$.

a. 4/5

b. 3/5

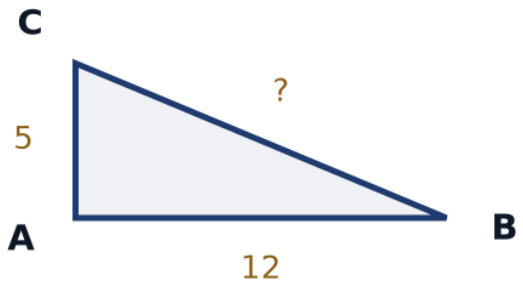
c. 1

d. 5/3

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40 [Challenging] • Target: MCQ — Solving basic trigonometric equations • Cross-Lesson: 3.01 Sequences & Series

■■ ■■■■■ ■■■■■ ■■■■■■■■■ ■■■■ ■■■■■■■■ ■■■■■■■■■ 5 ■ 12. ■■ ■■■■ ■■■■■■■ In a right triangle, the two legs have lengths 5 and 12. Find the hypotenuse.



a. 12

b. 9

c. 15

d. 13

41 [Challenging] • Target: MCQ — Reference angles in other quadrants • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

■■ ■■■■■■■ ABC■ ■■■■ ■■■■ $\angle A = 45^\circ$ ■ $\angle B = 60^\circ$ ■ ■■■■■ ■■■■■■ $a = 10$■ ■■■■ ■■ ■■■■ ■■■■■■ b ■■■■■■ In triangle ABC, if $\angle A = 45^\circ$, $\angle B = 60^\circ$, and the length of side $a = 10$, what is the exact length of side b?

a. $\frac{10\sqrt{3}}{\sqrt{2}}$

b. $\frac{10\sqrt{2}}{\sqrt{3}}$

c. $\frac{10\sqrt{6}}{2}$

d. $\frac{10\sqrt{3}}{2}$

42 [Challenging] • Target: MCQ — Complementary angle relationships • Cross-Lesson: 6.01 Limits & Derivatives

■■ ■■■■■■ ■■■■■■■■ ■■■■■■ ■■■■■ ABC ■■■■ $AB = 10$ ■ $AC = 15$. ■■■■■ ■■■■■■■ ■■■■■■■■■■ AD ■■■■■■■■ BAC■ ■■■■ ■■■■ D ■■■■ BC. ■■■■ ■■■■ $BD = 6$ ■■■■■ $BC = x$■ ■■■■■■ ■■■■■ x. In the attached figure, we have triangle ABC where $AB = 10$ and $AC = 15$. The line segment AD bisects angle BAC, with D lying on BC. If $BD = 6$ and the length of $BC = x$, find the value of x.

a. $x = 8$

b. $x = 9$

c. $x = 10$

d. $x = 12$

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43 [Challenging] • Target: MCQ — The unit circle: coordinates of key points • Cross-Lesson: 7.01 Integration & Applications

■■ ■■■■■ ■■■■■■■ ■■■■■ ■■■■ ABC ■■■ $AB = 10$ ■■■ $BC = 15$ ■■■ ■■■■■ ■■■■■■■ $B = 60^\circ$. ■■■ ■■ ■■■ ■■■■■■ ■■ ■■■■■ A ■■■ ■■■■■ BC ■■■■■ BC ■■ ■■■■■■ D■ ■■■■■ ■■■ ■■■■■ AD. In the shown figure, we have a triangle ABC where $AB = 10$ cm, $BC = 15$ cm, and the measure of angle $B = 60^\circ$. If an altitude is drawn from vertex A to side BC, intersecting BC at point D, find the length of side AD.

a. $5\sqrt{3}$ ■■

b. $10\sqrt{3}$ ■■

c. $7.5\sqrt{3}$ ■■

d. $15\sqrt{3}$ ■■

44 [Challenging] • Target: MCQ — Word problems: angles of elevation • Cross-Lesson: 8.01 Exponentials & Logarithms

■■ ■■■■■ ■■■■■■■■ ■■■■■■ ABC ■■ ■■■■■ ■■■■■ $AB = 10$■ $AC = 15$■ ■■■■■■■ ■■■■■■■■ AD ■■■■ ■■■■■■■ A■ ■■■ D ■■■ ■■■ BC. ■■■ ■■■ ■■■ $BD = 6$■ ■■■■■ ■■■ DC. In the given figure, triangle ABC has side lengths $AB = 10$, $AC = 15$, and segment AD bisects angle A, with D on BC. If the length of $BD = 6$, find the length of DC.

a. 8

b. 9

c. 10

d. 12

45 [Challenging] • Target: Right-triangle trigonometric ratios • Cross-Lesson: 9.01 Statistics & Probability

■■ ■■■■ ABC ■ ■■■ ■■■ $a = 5$■ $b = 7$■ ■■■■■■■■ $C = 60^\circ$ ■ ■■■ ■■■■ ■■■ ■■■■■ c■

In triangle ABC, if $a = 5$, $b = 7$, and angle $C = 60^\circ$, what is the length of side c?

a. $\sqrt{39}$

b. $\sqrt{49}$

c. $\sqrt{61}$

d. $\sqrt{75}$

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